SIVA KUMAR ANNAM | Python Developer

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PROFESSIONAL SUMMARY

Python Developer with 1.5+ years of hands-on experience building automation scripts, RESTful APIs, and data-processing tools for network monitoring and service assurance platforms. Skilled in Python, Flask, Perl, Bash, and database migrations (MySQL → PostgreSQL). Delivered automation that reduced manual effort by ~30%. Additional hands-on experience with Generative AI — built a RAG chatbot using LangChain and Ollama. Strong focus on clean code, performance optimization, API design, and continuous learning in DevOps, cloud and modern AI technologies.

WORK EXPERIENCE

Associate Software Engineer | Python Developer

MasterCom Technology Services India Pvt Ltd, Bangalore | Nov 2024 – Present (~1.5 years)

Project: SevOne NMS & Service Assurance Automation | Client: Tata Communications Ltd (Pune)

- Automated data collection, reporting, and backups using Python/Perl scripts, achieving a 35% reduction in manual effort for network operations.
- Engineered Flask RESTful APIs, streamlining business logic and caching to reduce database strain by 30%.
- Optimized complex SQL join logic during migration, decreasing average query execution time by 2.5 seconds.
- Improved efficient database query strategies for MySQL and PostgreSQL, leading to a 30% reduction in database strain and improved application performance metrics.
- Orchestrated daily SevOne dashboard generation, correlating 5,000+ SNMP metrics and NetFlow flows, reducing average incident detection time from 45 to 20 minutes
- Stabilized network performance by swiftly addressing service degradation issues, leading to a 15% reduction in average incident resolution time and preventing potential outages.
- Diagnosed and resolved complex technical issues across Assure1 & Spotlight platforms, reducing incident resolution time by 15% through root cause analysis.
- Diagnosed and resolved 10 recurring technical issues, implementing documented solutions that prevented future escalations and reduced incident volume by 20%.
- Wrote clean, maintainable code and added basic unit tests with Pytest to improve code reliability.
- Mentored new team members on Python automation and SevOne NMS practices, helping them become productive faster.

PROJECTS

1. SevOne NMS & Service Assurance Automation (Client: Tata Communications Ltd)

Technologies: Python, Perl, Bash, Flask, MySQL, PostgreSQL, SevOne NMS, DISHA (Assure1), Linux, ServiceNow

- Delivered end-to-end automation for monitoring, reporting, and operational tasks on SevOne NMS, reducing manual effort by 40% and accelerating data insights.
- Developed Python/Perl scripts to automate metrics collection and report generation, accelerating data insights by 15% for a large telecom client.
- Completed migration of vital database workloads from MySQL to PostgreSQL, optimizing joins by 30% and ensuring full data validation.

2. RAG Chatbot using LangChain + Ollama (Hands-on Generative AI Project)

Technologies: Python, LangChain, Ollama, LLMs, Embeddings, RAG, Prompt Engineering | GitHub: github.com/Sivakumarannam/rag-chatbot-langchain-ollama

- Architected a RAG chatbot utilizing LangChain and local Ollama LLMs, achieving a 15% reduction in hallucination rates and improving answer precision.
- Implemented document loading, text splitting, embedding generation, and vector store for accurate context-aware responses.
- Refined prompt engineering strategies to improve answer quality and minimize hallucinations.
- Created a production-ready chatbot that answers questions from custom knowledge bases (PDF/text documents).
- Applied key Generative AI concepts including LLMs, Embeddings, Fine-tuning, RAG, and Agentic AI patterns.
- Set up the full development environment, integrated ChatGPT API for comparison, and documented the complete workflow.

TECHNICAL SKILLS

Programming & Scripting: Python (strong), Perl, Bash, Shell Scripting, JavaScript, PL/SQL
Web & Frameworks: Flask, Django (basic knowledge)
Generative AI / LLM: LangChain, Ollama, RAG, Prompt Engineering, Embeddings, ChatGPT API, LLMs
Cloud & DevOps: AWS (EC2, S3, RDS, VPC, IAM, Lambda, CloudWatch, CodePipeline, EKS, ECS), Terraform, Ansible, Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Nexus, SonarQube
Monitoring & Observability: SevOne NMS, PMS, DISHA (Assure1), Prometheus, Grafana, CloudWatch
Databases: MySQL, PostgreSQL, DBeaver — CRUD operations, migrations, query optimization
Networking & Protocols: SNMP, NetFlow, performance and alert log analysis, Linux/Unix administration
Version Control & ITSM: Git, ServiceNow, GitOps practices
Other: Pytest, VS Code, PuTTY, WinSCP, writing clean and maintainable code

KEY LEARNING & SELF-PROJECTS

GitOps Deployment Platform (Capstone) — Built an end-to-end GitOps-style platform using GitHub Actions and Kubernetes for declarative continuous delivery. Focused on pipeline design, environment promotion and automated deployments.

DevOps & Cloud Mastery (Self-Paced) — Linux & server management, Networking fundamentals, YAML/JSON/Bash, AWS core services (IAM, EC2, S3, RDS, VPC, Lambda, Auto Scaling, Route 53, Systems Manager), Terraform modules & backends, Ansible, Prometheus/Grafana/Loki, Docker & production-grade Kubernetes, full CI/CD with Jenkins/GitHub Actions/GitLab, Nexus, SonarQube, Slack integration. Python scripting for automation and AWS tasks.

Generative AI Learning Path — Structured modules on Generative AI fundamentals, LLMs, Prompt Engineering, Embeddings, Fine-tuning, RAG, Agentic AI, industry use-cases, ethical considerations, and practical chatbot development with ChatGPT API and local LLMs (Ollama).

CERTIFICATIONS

- Python Full Stack Development — 2023
- Full Stack Development Internship — Innomatics Research Labs, 2023

EDUCATION

Bachelor of Technology in Computer Science and Engineering
St. Mary's Group of Institutions, Guntur, Andhra Pradesh | 2019 – 2023 | CGPA – 6.8

Intermediate (Class 12 / +2)
NRI Junior College, Tenali, Andhra Pradesh| 2017 – 2019 | CGPA: 8.7

Secondary School Certificate (Class 10)
Nehru Niketan English Medium High School |Tenali, Andhra Pradesh| CGPA: 8.7

LANGUAGES

English, Telugu (Mother Tongue), Hindi

STRENGTHS

- Strong focus on automation, monitoring, and reducing manual operational effort
- Experience collaborating with cross-functional teams and mentoring juniors
- Consistent attention to SLA compliance, documentation, and knowledge sharing
- Active learner in Generative AI, cloud, and scalable application development